

# **F24-065-D-SceneCraft**

Project Team

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# Chapter 1

## Introduction

This document defines the requirements for SceneCraft an application that will revolutionize the conversion of book manuscripts to screenplay. SceneCraft's goal is to simplify adaptation for writers by offering them an intuitive tool that takes manuscripts as input and provides output in the form of editable and industry-standard screenplays.

This requirement document is therefore targeted at stakeholders involved in the development and use of SceneCraft:

1. **Developers:** To comprehend the technical specifications and functional requirements for development.
2. **Project Managers:** To plan and control the development process, for managing resources and evaluating the results.
3. **Marketing Staff:** To understand the product features and the enhancements that may be done for promotional purposes.
4. **Users (Book Authors, Screenwriters, Producers):** To get familiar with key features of the app and ways that it will be helpful to their creativity.
5. **Testers:** To write and execute test cases concerning the defined requirements.
6. **Documentation Writers:** To develop user manuals and help documentation according to the system's capabilities and options.

### 1.1 Problem Statement

When it comes to cinema, most movies created are adaptations of novels, and the role of bringing this book into an entirely new medium falls to a movie. A major component of

this effort is the translation of book manuscripts into screenplays. This calls for a comprehension of all the works visually depicted in the book, the entire story arc of the books, and how precisely the characters advance the plot. This task is very time-consuming and normally it may take most writers several months to come up with these screenplays. The other problem, which writers experience while working on a movie script, is tagging the script components. The screenplay tags enhance the script by underlining the screenplay's aspects such as the characters' names, the props, VFX, and others. Current screenplay tools used require manual handling for such tasks since they are repetitive thus slowing the scripting.

## 1.2 Scope

As a new tool developed to help make the conversion of book manuscripts to screenplay easier, SceneCraft is a powerful tool for any writer. This project also includes the creation of an effective and intuitive interface where book manuscripts may be uploaded and where the program sends out properly formatted, industry-standard screenplays that users can edit directly as part of the project scope. Elements like plot points, characters, and settings from the story will be determined by our application. SceneCraft will then grow these elements into a classic screenplay form from which the heart of the story will be retained though it will be changed and put into a new visual narrative media form. The system will also feature an editing tool that enables the user to edit the screenplay, suggesting improvements from artificial intelligence. An automated feature in SceneCraft will be added to highlight and tag key production elements within the Screenplay. The goal of this project is to help increase the availability of opportunities where creative people have the chance to apply their talents toward a book-to-screen project. SceneCraft will enable the adaptation and editing tool but will not extend through the other stages of the filmmaking process. Our goal is to bring people closer to where writing and screenplay visuals meet, allowing those who want to tell stories to get them on the screen more easily than before.

## 1.3 Modules

### 1.3.1 Module 1

In this module, we will establish the foundation on which subsequent stages of the project will be built.

#### 1. Dataset Creation

2. Tagging Model Training
3. Script Editor Interface Development
4. Book Scraping

### **1.3.2 Module 2**

This module aims to leverage AI with Natural Language Processing and Generative AI techniques for our specific screenplay purposes.

1. Entity Extraction
2. Prompt Engineering
3. LLM fine-tuning
4. RAG system implementation

### **1.3.3 Module 3**

This module aims to implement and enhance the front end of the system to better support an environment for creating a screenplay.

1. Format Conversion
2. Real-time editing
3. AI-assisted suggestions
4. Integration

### **1.3.4 Module 4**

The last module is spent on fine-tuning and enhancing the performance of the system for a better experience for the user.

1. Tagging Model fine-tuning
2. Testing
3. UI/UX enhancement



## 1.4 User Classes and Characteristics

Table 1.1: User Classes and Characteristics

| User class          | Description                                                                                                                                                                                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Book Authors        | Book Writers use SceneCraft’s features to create and edit their manuscripts. With chapter organization tools and AI assistance, writers can focus on long-form narrative prose. Many authors will also be interested in converting from book to screenplay format, they may have limited experience with it. |
| Screenwriters       | Screenwriters mainly use the screenplay writing feature. They are familiar with the industry standards and will utilize the screenplay formatting tools and scene structuring features. Screenwriters may also use the book-to-screenplay conversion tool when adapting existing work.                       |
| Assistant Producers | Assistant Producers oversee the screen-writing process. They assess the adaptations done after converting from book to screenplay format and then provide feedback. They will also make use of an automated element tagging feature for budgeting which would otherwise take a lot of time                   |
| Aspiring Writers    | Aspiring Writers are likely to use all three of the main features. They may start with writing a book to develop their narrative skills. They would then proceed to convert it into screenplay format and then use the screenplay writing features to improve it further.                                    |

# Chapter 2

## Project Requirements

This chapter describes the functional and non-functional requirements of the project.

### 2.1 Use-case Diagram

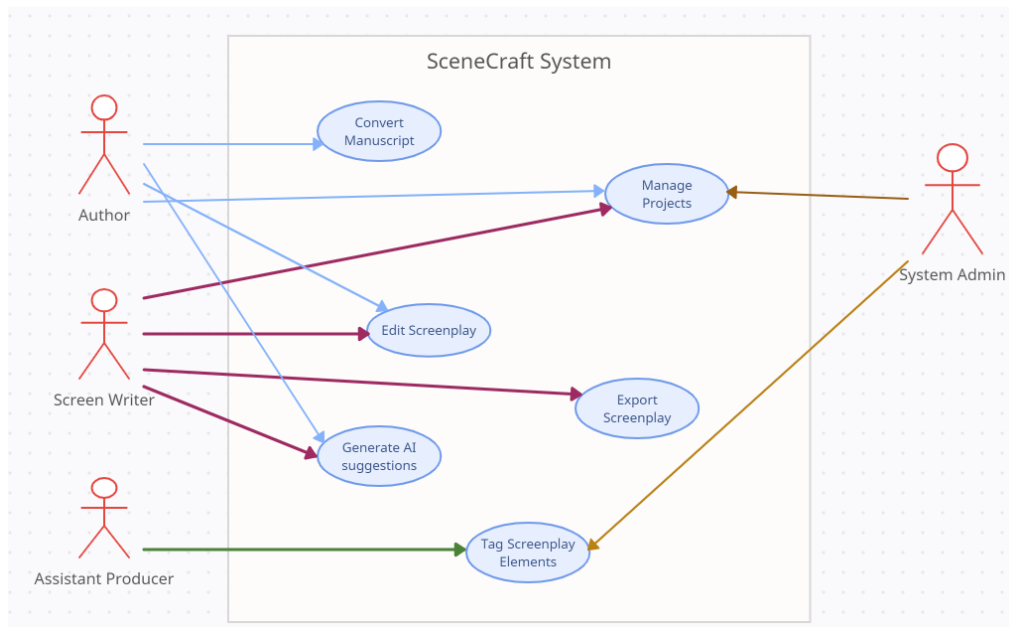


Figure 2.1: SceneCraft Use Case Diagram

## **2.2 Functional Requirements**

### **2.2.1 Module 1**

The following are the functional requirements for Module 1:

1. FR1: The SceneCraft system shall implement a tagging model which will identify and label key elements (e.g., characters, settings, dialogue) in a screenplay.
2. FR2: The SceneCraft system shall provide users with a script editor interface that will allow users to edit and format screenplays.
3. FR3: The SceneCraft system shall implement a book scrapping functionality that can extract content from EPUB files. The system shall process and extract a standard novel-length book while maintaining essential metadata such as title, author and publication date.

### **2.2.2 Module 2**

The following are the functional requirements for Module 2:

1. FR1: The SceneCraft system shall automatically extract entities (such as characters, locations, and key plot points) from uploaded book manuscripts.

### **2.2.3 Module 3**

The following are the functional requirements for Module 3:

1. FR1: The SceneCraft system shall transform book manuscript text into a neatly organized screenplay framework that complies with industry norms and correctly recognizes scene headings, action lines, character names, and dialogues.
2. FR2: The SceneCraft system shall create a real-time editing interface that enables users to edit the scripted text while seeing the changes made by multiple users on the same document flawlessly.
3. FR3: The SceneCraft system shall offer recommendations from AI for better dialogue and character growth while editing scenes that are relevant to users according to their input and created upon triggering the suggestion feature.

4. FR4: The SceneCraft system shall merge the features for format conversion and real-time editing with AI suggestions into a consistent user interface so users can access major features in minimum clicks.

### **2.2.4 Module 4**

The following are the functional requirements for Module 4:

1. FR1: The SceneCraft system shall optimize its tagging model to increase accuracy in tagging screenplay details (for instance scene headings and dialogue).
2. FR2: The SceneCraft system shall review extensive testing that includes unit testing and integration tests along with user acceptance tests with all critical bugs fixed before the end of development.

## **2.3 Non-Functional Requirements**

### **2.3.1 Reliability**

- REL-1: The SceneCraft system shall sustain an MTBF greater than 720 hours (30 days). A disruption happens if it stops users from engaging with their projects or completing essential actions within 5 minutes.
- REL-2: The SceneCraft system shall set up daily automated backups for every project and system configuration that permit users to retrieve from any backup point within 30 minutes.
- REL-3: The SceneCraft system shall constantly assess its well-being and alert the team for any unexpected issues or signs of trouble. Within 5 minutes of any detected issues occurring, the maintenance team will receive alerts.

### **2.3.2 Usability**

- USE-1: The SceneCraft system shall allow users to change a manuscript to a screenplay structure within 5 clicks from the main interface.
- USE-2: The SceneCraft system shall offer assistance relevant to each core feature with a single click or tap from every page.
- USE-3: The SceneCraft system shall save user updates automatically every 2 minutes and let users return to any older version with just two clicks.

USE-4: The SceneCraft system's UI shall function as an easy-to-navigate interface supporting keyboard shortcuts with a detailed shortcut overview reachable at any moment through pressing F1.

USE-5: The SceneCraft system shall show direct and clear error notifications for all input mistakes alongside suggested corrections in under 5 seconds.

### **2.3.3 Performance**

PER-1: The SceneCraft system shall provide its AI recommendations for manuscripts within 5 seconds of user input.

PER-2: The SceneCraft editor shall facilitate real-time team editing with no more than 500 milliseconds for users connected on a 50 Mbps or higher Internet connection.

PER-3: The SceneCraft system shall produce a PDF version of the completed screenplay in less than 10 seconds for files up to 200 pages.

### **2.3.4 Security**

SEC-1: The SceneCraft system shall not allow unauthorized access attempts by people with highly sophisticated computer skills and knowledge of common hacking techniques the majority of the time.

SEC-2: The SceneCraft protocol shall protect all data transmissions from the SceneCraft server to client applications.

SEC-3: SceneCraft's user authentication mechanisms shall be able to resist common attack vectors (brute force, dictionary attacks, and credential stuffing) with a success rate of over 50%.

SEC-4: The SceneCraft system shall protect stored user data so that the physical hardware of the system would have to be breached by a team of skilled computer forensics experts, to extract any useful information from it.

## **2.4 Domain Model**

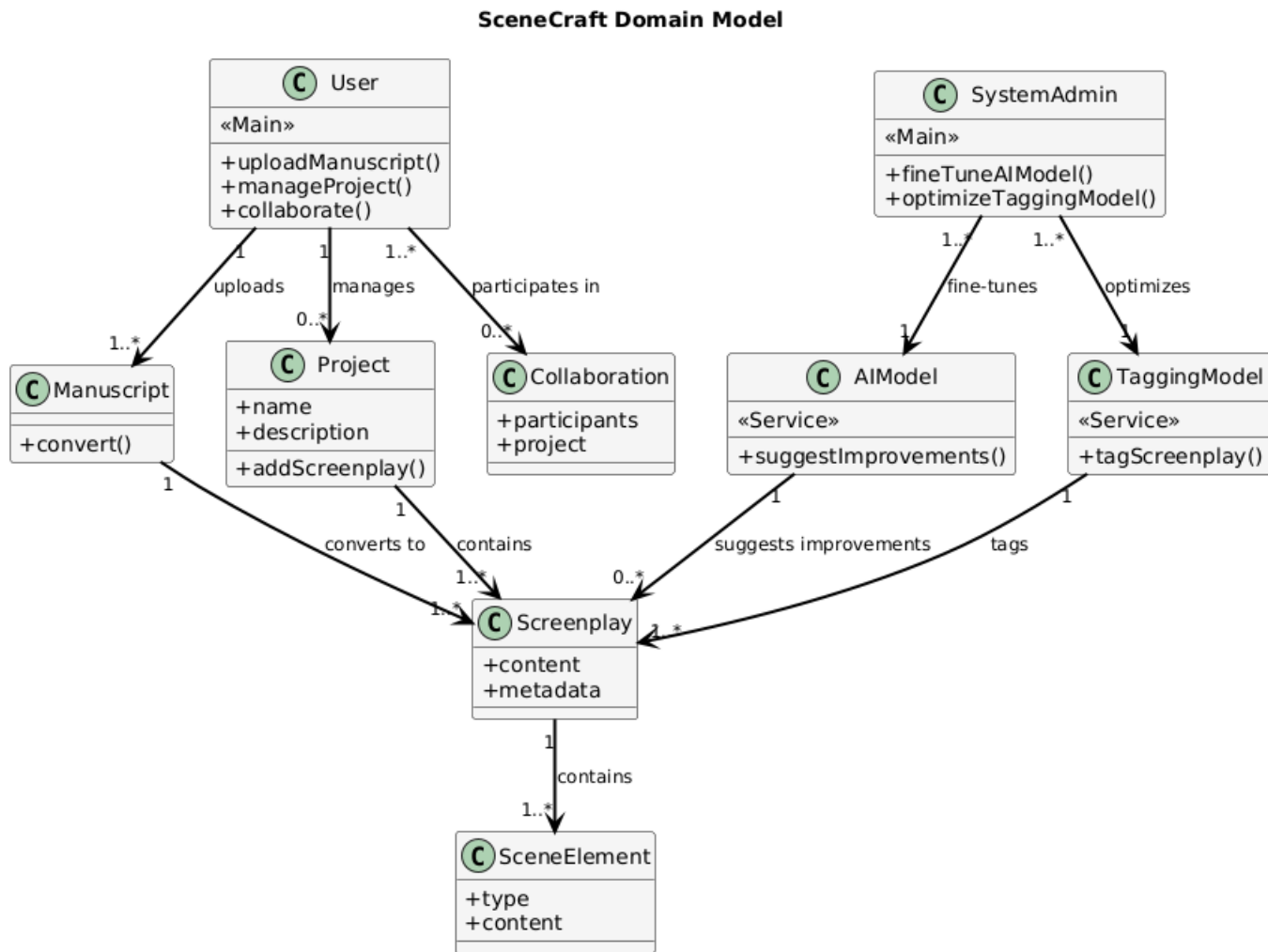


Figure 2.2: SceneCraft Domain Model

# Chapter 3

## System Overview

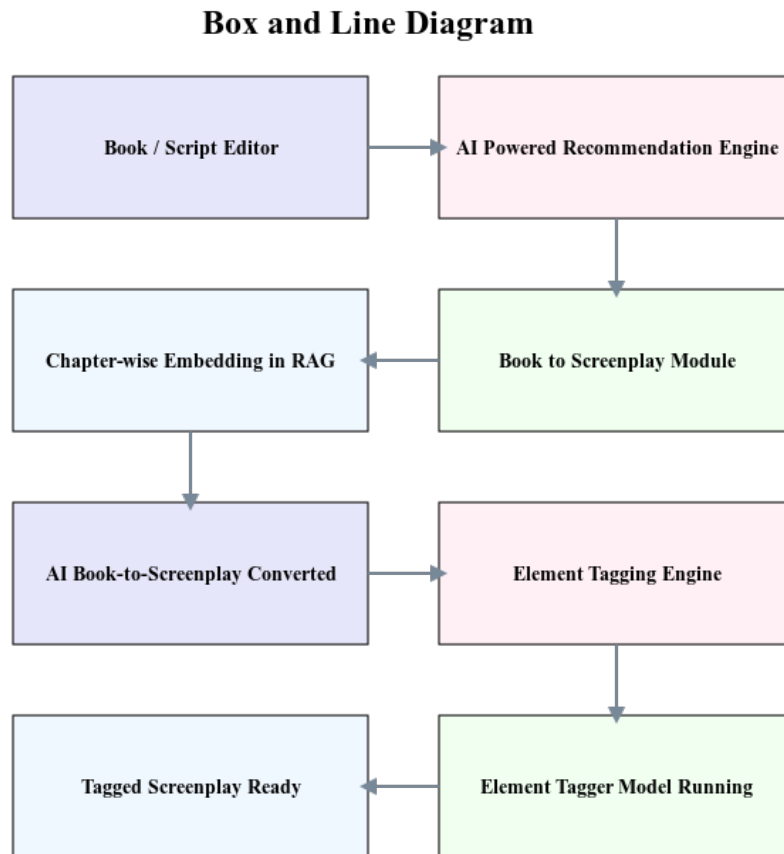


Figure 3.1: Box and Line Diagram

## 3.1 Architectural Design

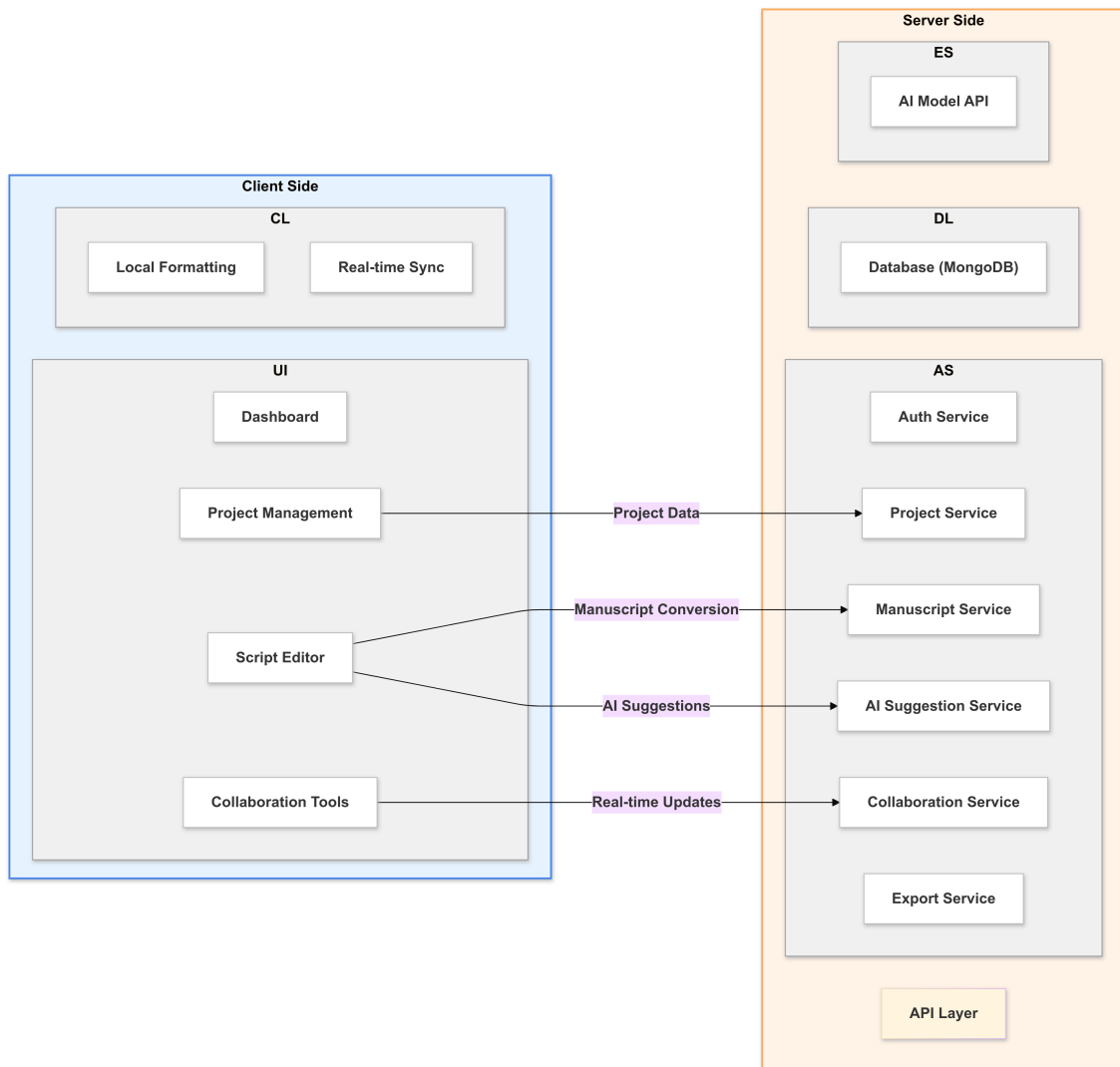


Figure 3.2: Architecture Diagram

The architecture diagram shows a system design for SceneCraft, divided into client-side and server-side components. Following is the modular structure and relationships between these components:

### 3.1.1 Client Side

#### 3.1.1.1 UI (User Interface)

- **Dashboard:** Offers an overview of projects and system status.



- **Project Management:** Handles creation and organization of screenplay projects.
- **Script Editor:** Editing and writing of screenplay with the main interface.
- **Collaboration Tools:** Allows real time collaboration with users.

#### **3.1.1.2 CL (Client Logic)**

- **Local Formatting:** Handles local text formatting and styling.
- **Real-time Sync:** Ensures that the data between client and server is synchronized.

### **3.1.2 Server Side**

#### **3.1.2.1 API Layer**

- Acts as the entry point for any client-server communications.

#### **3.1.2.2 AS (Application Services)**

- **Auth Service:** It manages user's authentication and authorization.
- **Project Service:** It handles all project related operations.
- **Manuscript Service:** Responsible for receiving manuscripts, and the subsequent conversion.
- **AI Suggestion Service:** Helps in writing by making suggestions to the document through artificial intelligence.
- **Aligns the use of the software to coordinate the activity of the people within the same project in real-time.**
- **Export Service:** Is responsible for the exportation of script to different media formats.

#### **3.1.2.3 DL (Data Layer)**

- **Database (MongoDB):** Stores persistent data to be used in application.

#### **3.1.2.4 ES (External Services)**

- **AI Model API:** Processes advanced information coming from external AI services.

### 3.1.3 Key Relationships and Workflows

1. Project Management in the UI communicates with the Project Service on the server side for project data operations.
2. The Script Editor interfaces with multiple server-side services:
  - Manuscript Conversion service
  - AI Suggestion Service for Intelligent Writing Assistance
3. Collaboration Tools interact with the Collaboration Service to facilitate real-time updates between users.
4. The server-side services (AS) interact with the Database (DL) for data persistence.
5. The Client Logic layer (CL) helps UI operations with local formats and real-integration, synchronization of time and functionality, making the use by the end users and applications more appealing.

## 3.2 Design Models

### 3.2.1 Activity Diagrams

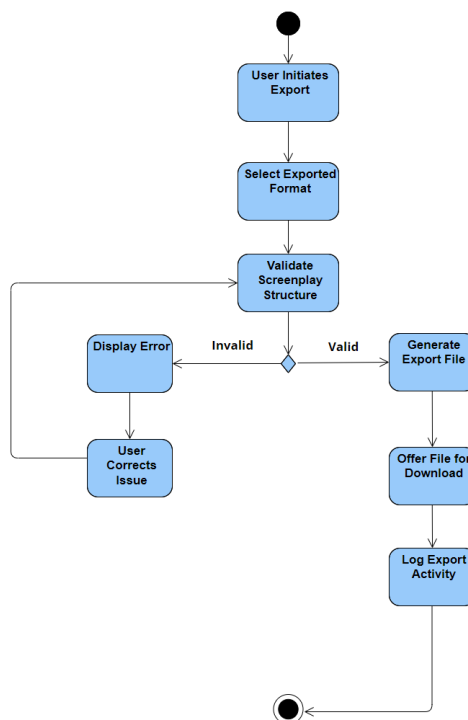


Figure 3.3: Export Screenplay Activity Diagram

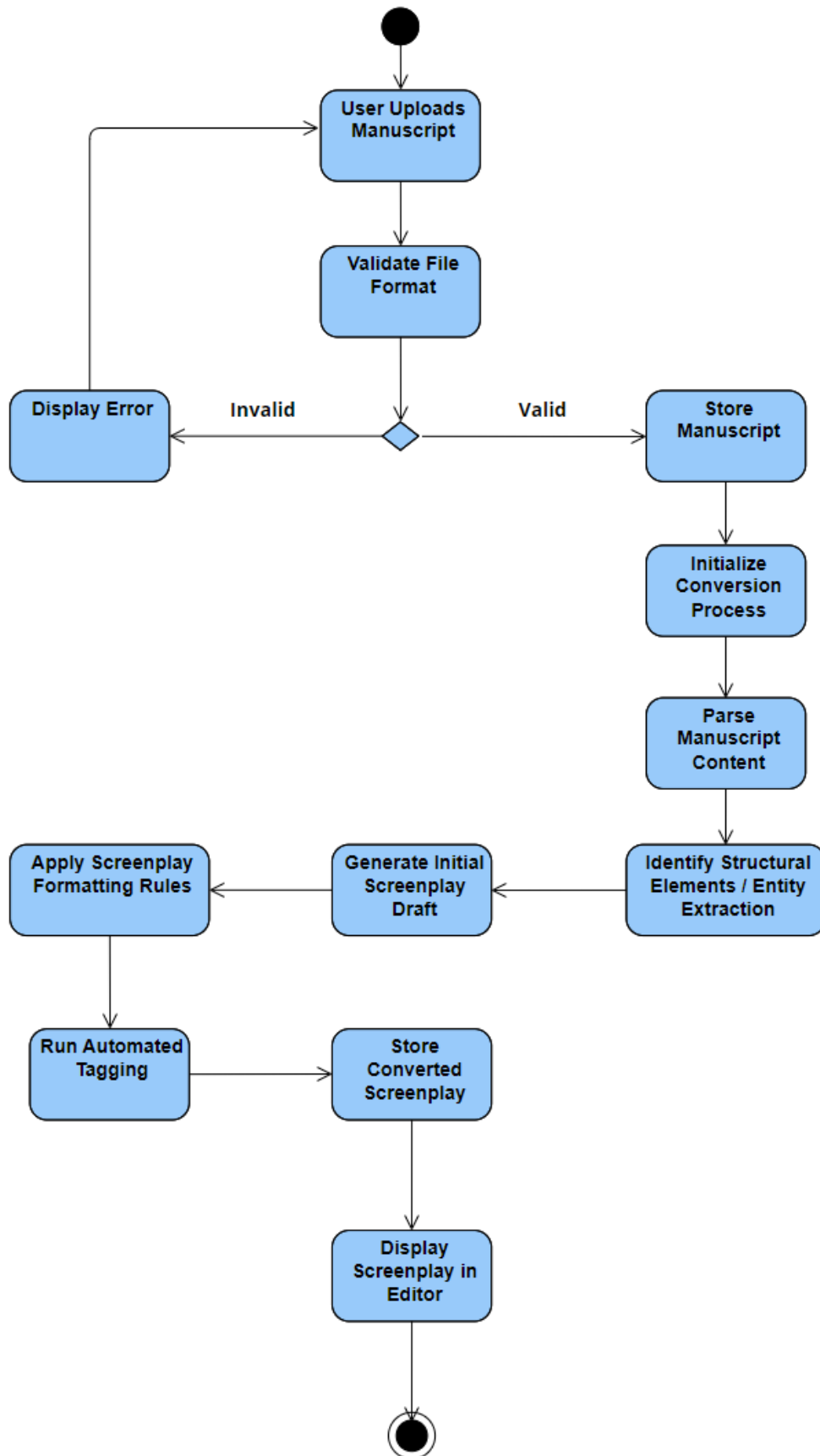


Figure 3.4: Book to Screenplay Conversion Activity Diagram

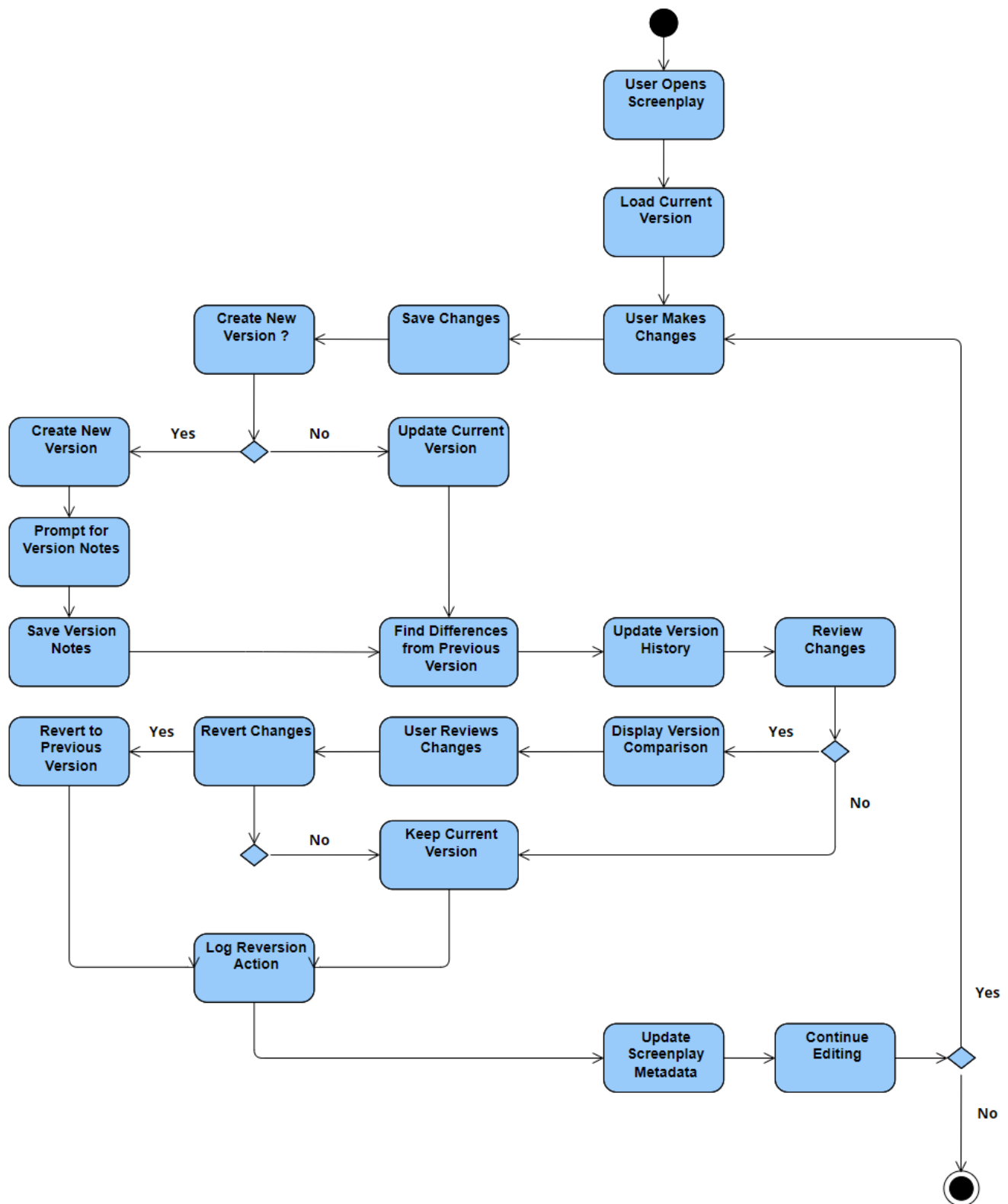


Figure 3.5: Modify Changes Activity Diagram

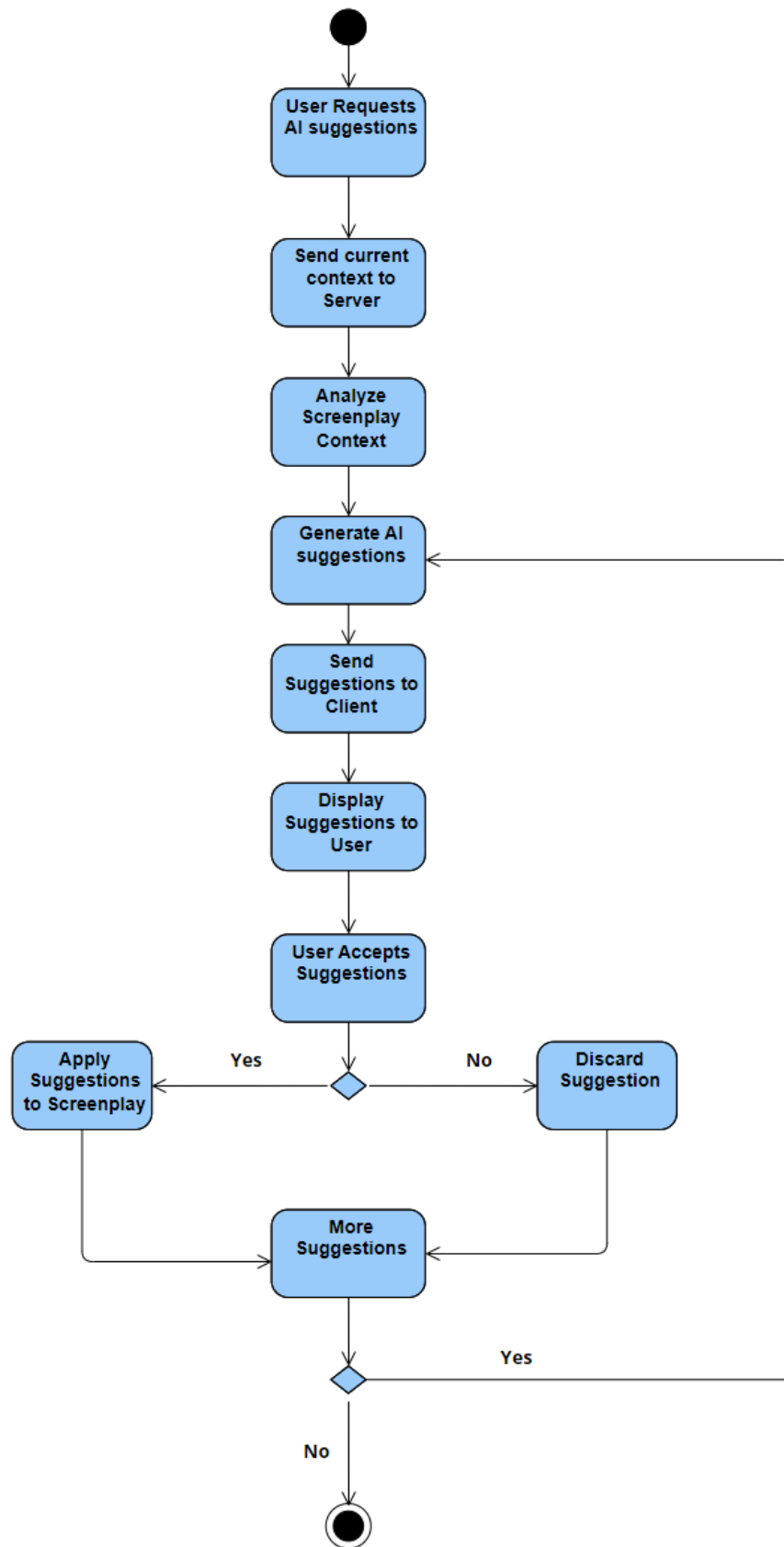


Figure 3.6: AI-assisted Suggestion Activity Diagram

### 3.2.2 Sequence Diagrams

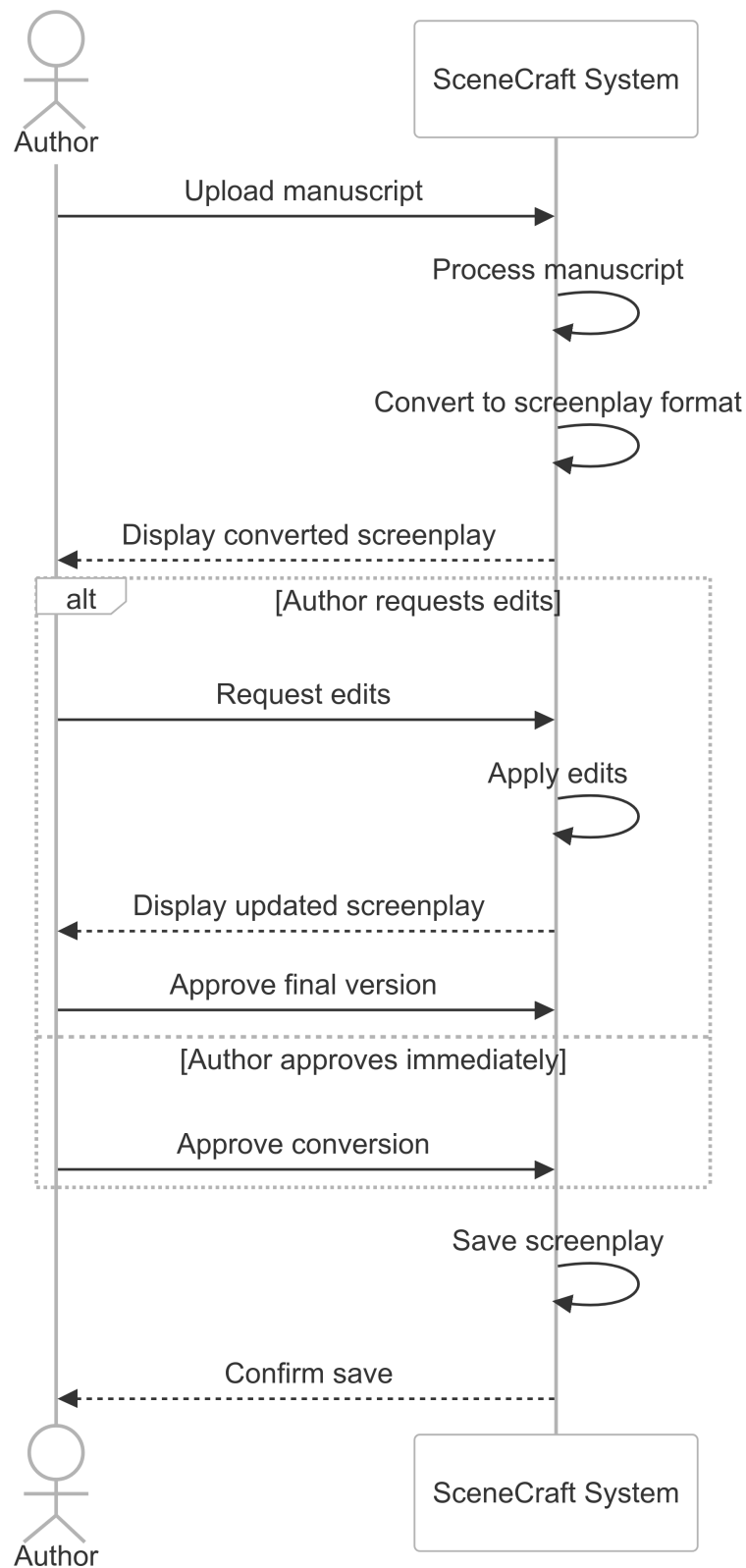


Figure 3.7: Manuscript Conversion Sequence Diagram

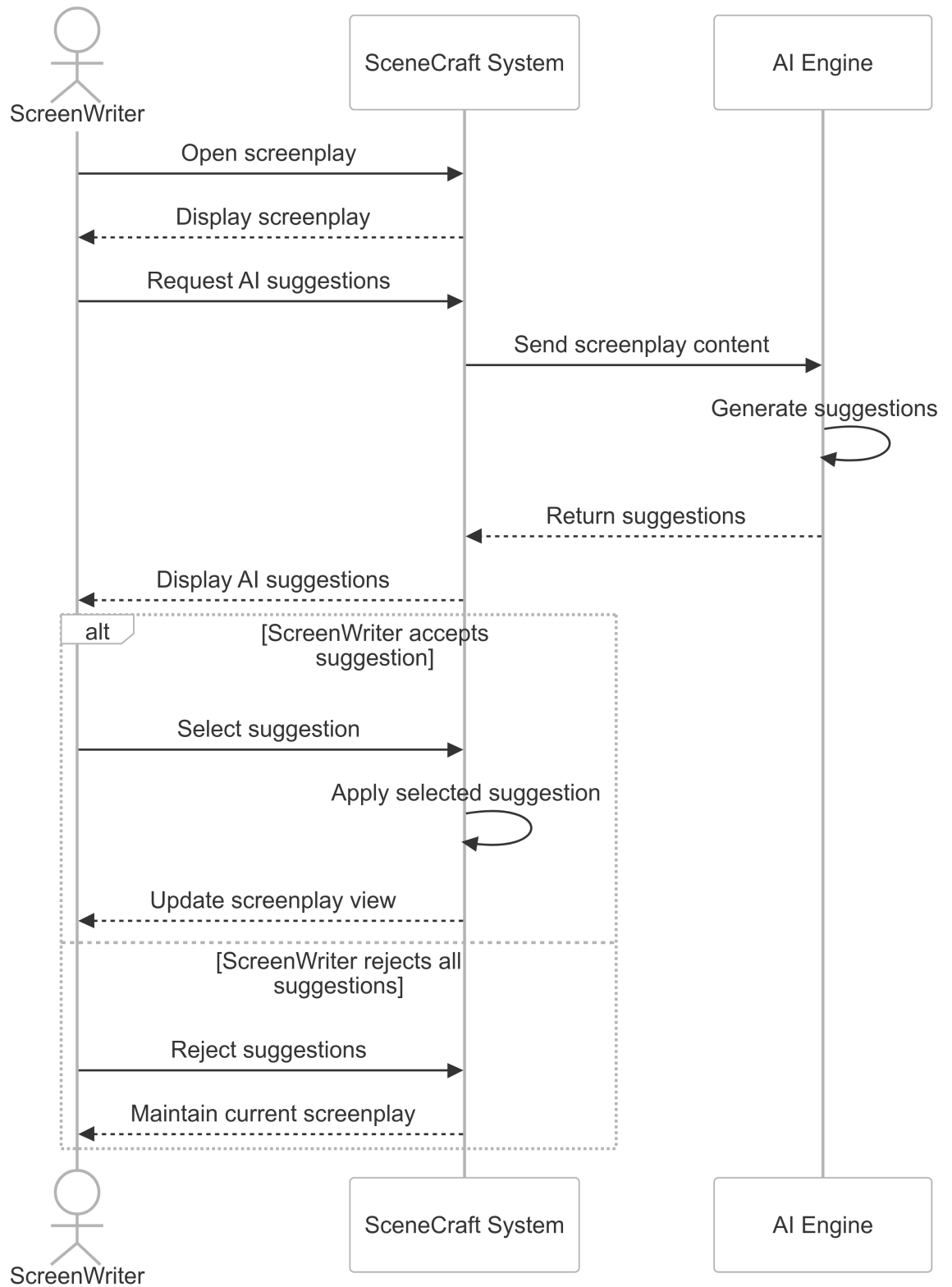


Figure 3.8: AI Suggestions Sequence Diagram

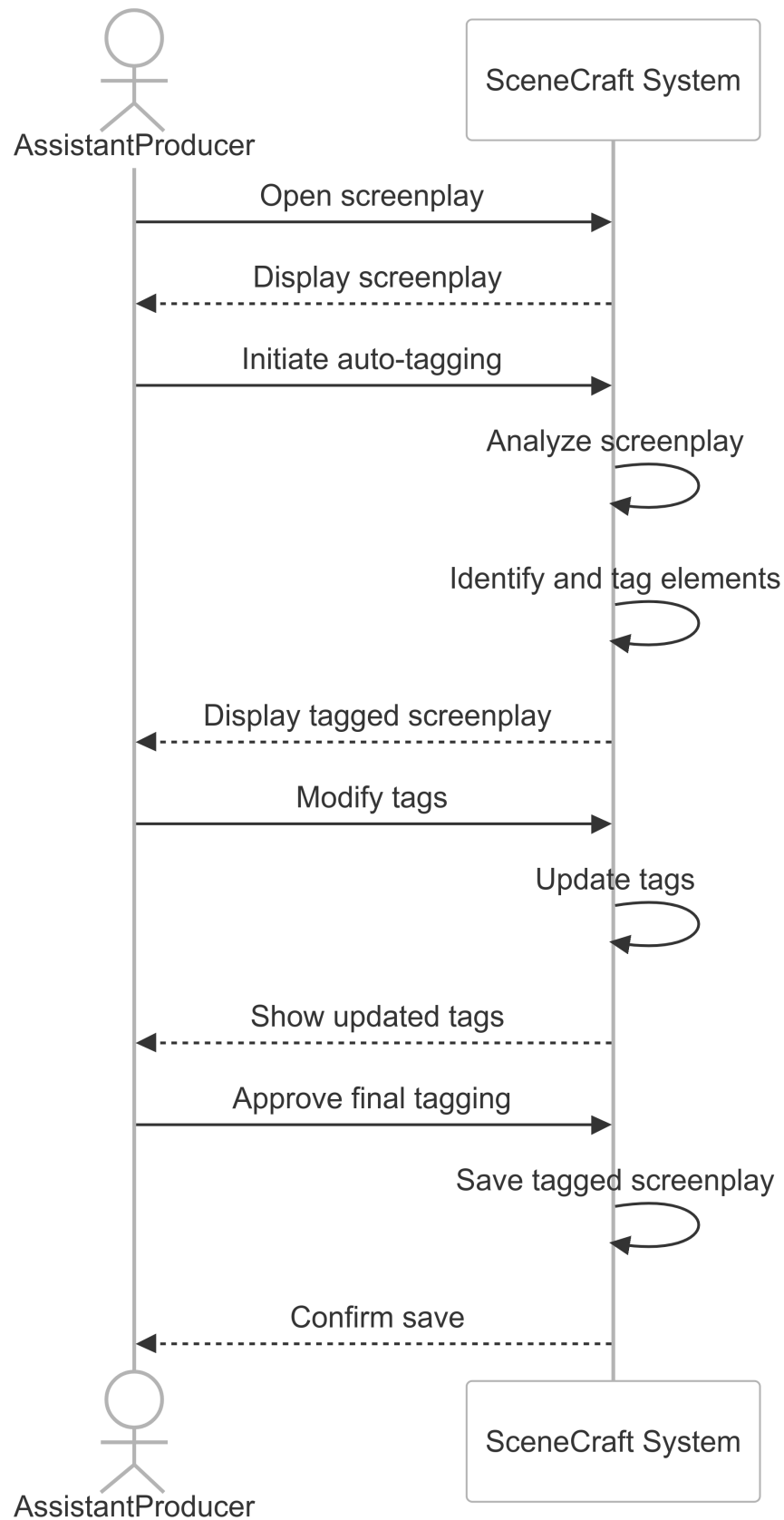


Figure 3.9: Screenplay Element Tagging Sequence Diagram



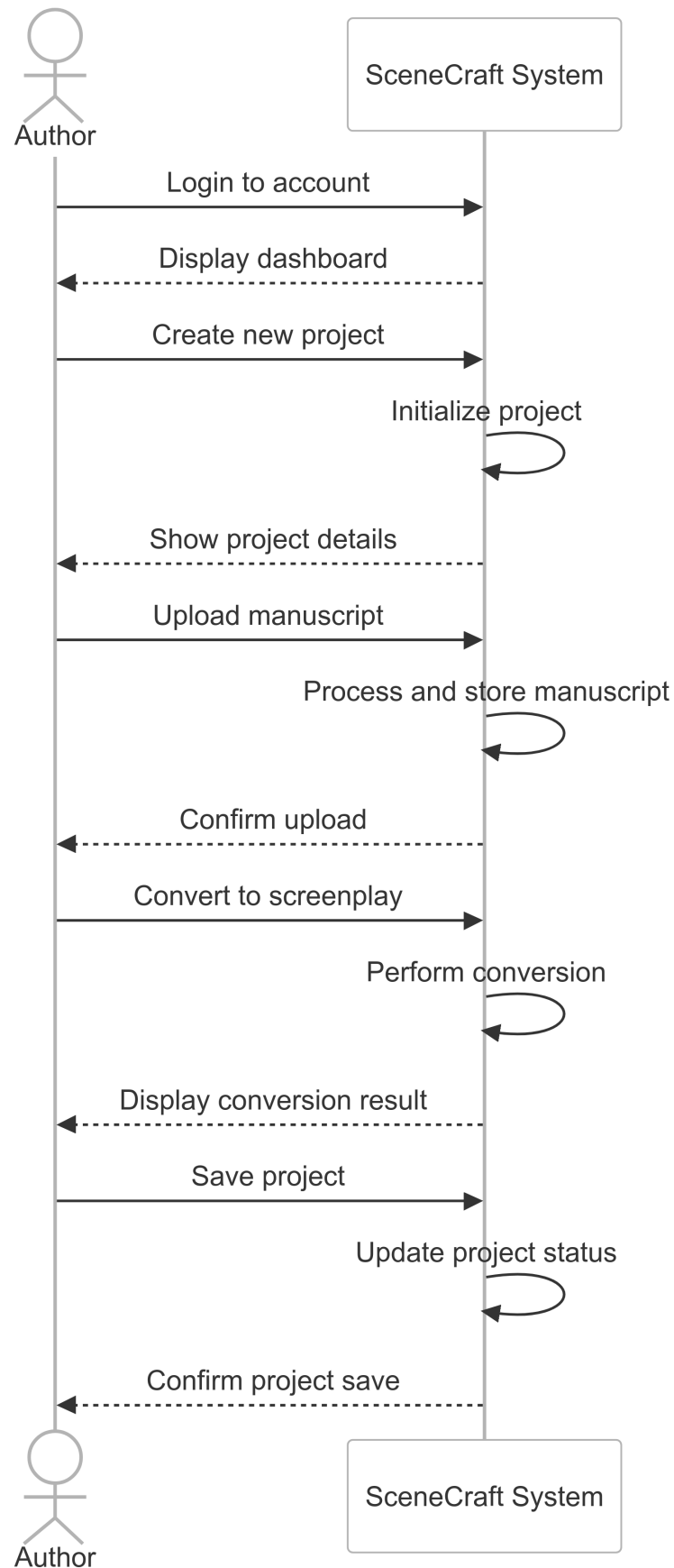


Figure 3.10: Project Management Sequence Diagram

### 3.2.3 Data flow Diagram

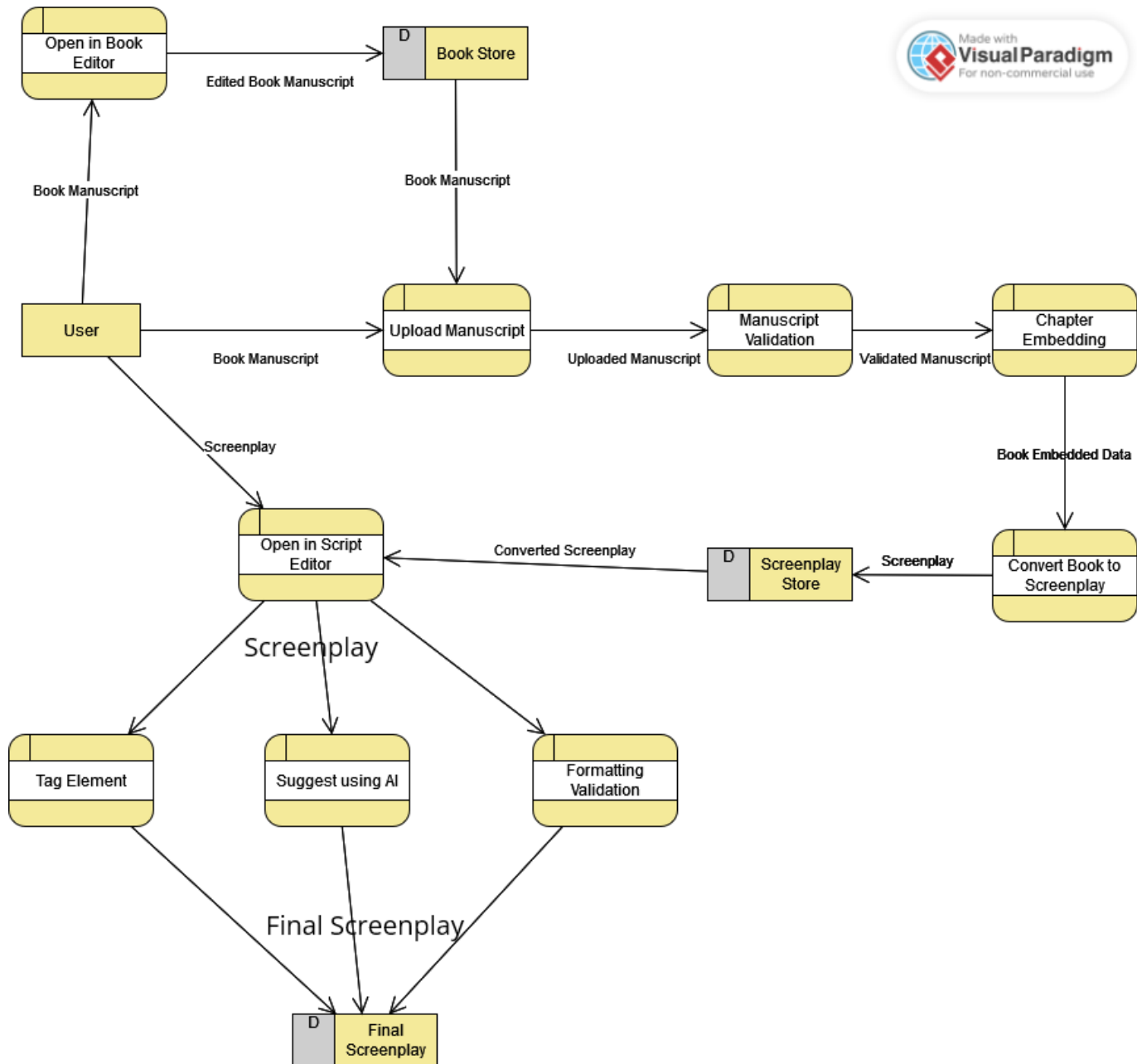


Figure 3.11: Data Flow Diagram

## 3.3 Data Design

Our data is stored in and organized with MongoDB, a NoSQL document-based database. It transforms the system's information into a collection of documents, corresponding to

major entities within the system. Here's how the data is structured and processed:

#### 3.3.1 Data Storage

1. We store each entity as a separate document for User, Project, Manuscript, Screenplay, Version, SceneElement, and AISuggestion in their respective MongoDB collections.
2. Entity relationships are maintained through reference fields (e.g. Screenplay document entity has a reference to a Project by the projectId field).

#### 3.3.2 Data Processing

1. When you upload a manuscript, it is added to the Manuscript collection and then processed to create a new Screenplay document.
2. When users edit the screenplay new Version documents are created to keep track of what has changed.
3. We extract SceneElements, storing them separately and linking them up with a Screenplay for quick retrieval and manipulation.
4. The AI suggestions are stored and linked to the relevant Screenplay, as separate documents.

#### 3.3.3 Data Organization

1. The top-level organizational unit is Projects that contain Manuscripts and Screenplays.
2. Version collection is used in screenplays for tracking changes in time.
3. Screenplays are linked to separate scene elements so they can be updated and rearranged quickly without altering the whole Screenplay document.

With this structure, the data can be managed flexibly and scale efficiently, including features such as collaborative editing, version control, and AI-assisted writing, while preserving the data integrity and making query and retrieval of information more efficient.

#### 3.3.4 ERD Diagram

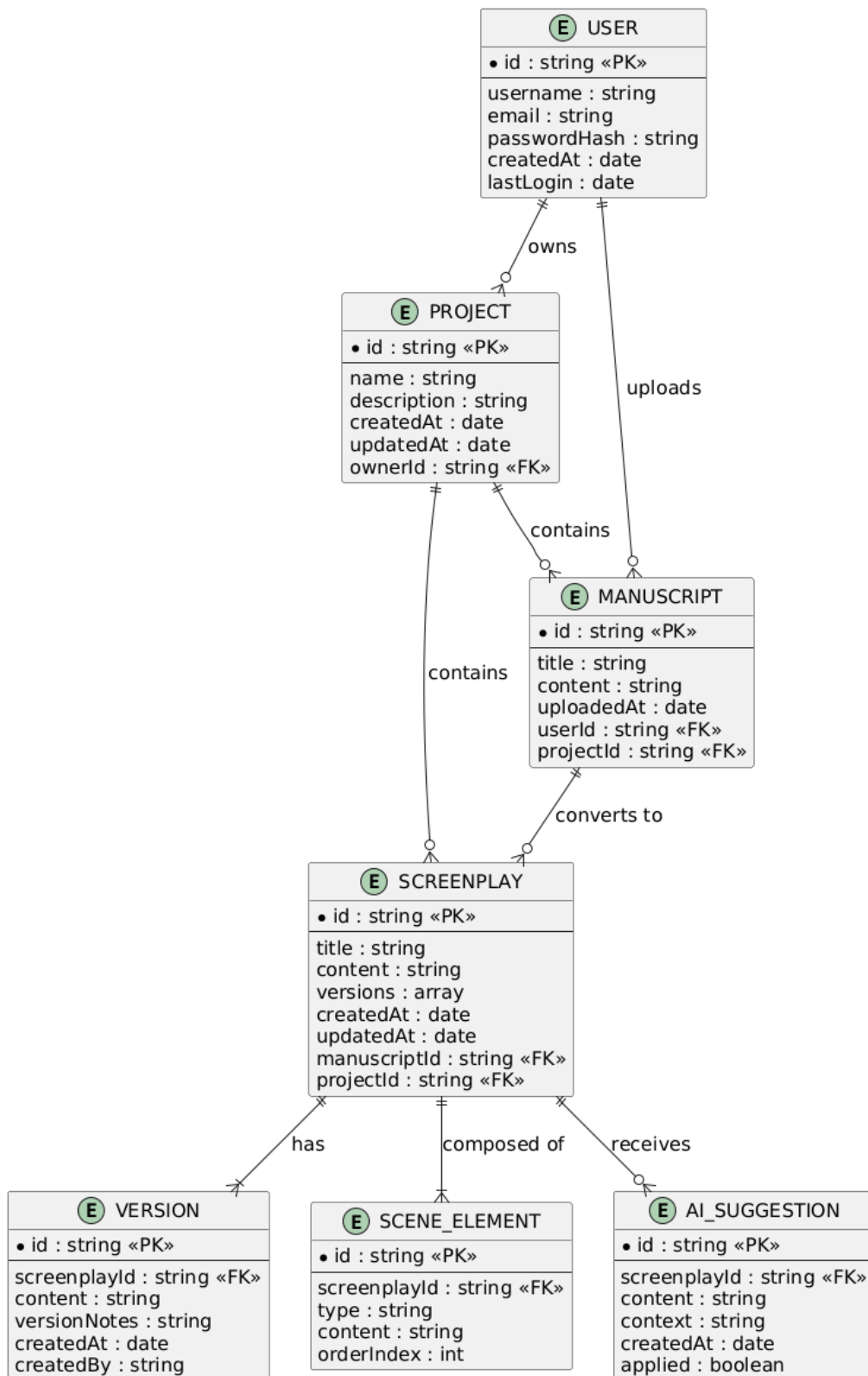


Figure 3.12: SceneCraft Entity-Relationship Diagram